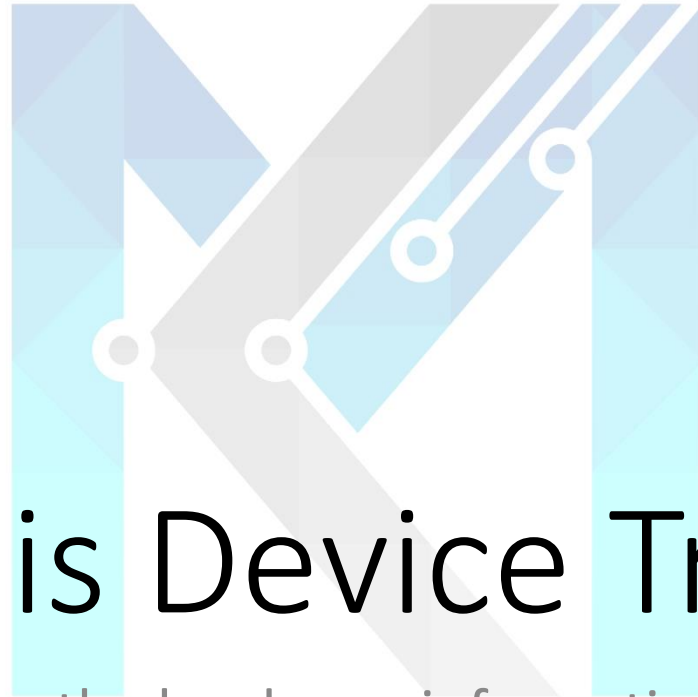




What is Device Tree?

How uboot pass the hardware information to the kernel?

Approach 1: ATAGS (Before the Device Tree)

The kernel contains the entire description of the hardware. The bootloader loads a single binary, the kernel image, and executes it. `ulmage` or `zlmage`.

a) ATAGS interface. Minimal information is passed from firmware to the kernel with a tagged list of predefined parameters.

`r0` : 0

`r1` : Machine type number

`r2` : Physical address of tagged list in system RAM

U-Boot command: `bootm <kernel img addr>`

Approach 2: (Booting with Device Tree)

The kernel no longer contains the description of the hardware, it is located in a separate binary: the **device tree blob (dtb)**.

The bootloader loads two binaries: the **kernel image** and the **DTB**.

Kernel image remains **ulmage** or **zimage**

Approach 2: (Booting with Device Tree)

Device Tree (DT), is a data structure and language for describing hardware.

More specifically, it is a description of hardware that is readable by an operating system so that the operating system doesn't need to hard code details of the machine.

DTB located in arch/arm/boot/dts, one per board

Approach 2: (Booting with Device Tree)

r0 : 0

r1 : Valid machine type number. When using a device tree, a single machine type number will often be assigned to represent a class or family of SoCs.

<http://www.arm.linux.org.uk/developer/machines/>

r2 : physical pointer to the device-tree block in RAM.

Device tree can be located anywhere in system RAM, but it should be aligned on a 64 bit boundary.

U-Boot command:

boot[mz] <kernel img addr> - <dtb addr>